Analysis of Scientific Data

Lecture 1

Course Overview

Evidence from Data

People

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  Drop-ins: Monday 3-4pm and Wednesday 2-3pm in 69-711

- Dr Christi Favor (School of Historical & Philosophical Inquiry)

  Ethics Lecturer – Module 2

- Associate Professor Simone Blomberg (School of the Environment)

  Experimental Design Lecturer – Module 10

Learning Objectives

1. Explain the nature of scientific data and the need for statistical analysis.
2. Identify factors related to the design of a scientific study, including sample size and power.
3. Identify and critically evaluate the role of data analysis and statistics in scientific research and publications.
4. Identify and critically evaluate ethical issues in scientific research.

Learning Objectives

5. Demonstrate a foundational knowledge of statistical methods by being able to carry out simple statistical procedures by hand.
6. Use statistical software appropriately and confidently for exploratory data analysis and to make relevant statistical conclusions.
7. Plan and carry out a research project, and analyse and communicate the results.

Learning Resources

- Course Profile
- Blackboard Site
- Echo360 Lecture Recordings
- UQ Extend
- Ed Discussion Board
- R (or RStudio)

Studying this course

At UQ you are expected to spend around 10 hours per week on a 2-unit course. For this course our intended time budget is roughly

1	Watching UQ Extend videos, taking notes
2	Attending lectures, taking notes
1	Reading UQ Extend text, trying questions
1.5	Participating in applied classes
1	Individual quiz questions
3.5	Working on assignments, engaging with Ed, etc.

Assessment Overview

Assessment details are available in the Course Profile and on Blackboard.

Groups

The Paper Review and Research Project can be completed in groups of up to three.

Register on Blackboard.

Ed Discussion

Click on the link in Blackboard to access the Ed Discussion site.

We will try to respond to *administrative questions* promptly but for *content questions* we will leave at least 24 hours for students to discuss first.

RStudio

Freely available for Mac, Windows and Linux.

Install **R** and then **RStudio**.

Download links are under Course Resources > RStudio on Blackboard.

Poll Question

Suppose a fair coin is tossed three times. What is the probability of obtaining three heads?

1. $\frac{1}{2}$
2. $\frac{1}{3}$
3. $\frac{1}{4}$
4. $\frac{1}{6}$
5. $\frac{1}{8}$

$\mathbb{P}(H) = \frac{1}{2}$

By independence

$$\mathbb{P}(HHH) = \left(\frac{1}{2}\right)^3 = \underline{\mathbb{P}}(H)\underline{\mathbb{P}}(H)\underline{\mathbb{P}}(H)$$

$$= \frac{1}{8}$$

Poll Question

How many hours of paid work have you done in the last 7 days?

1. None
2. 1-5
3. 6-10
4. 11-20
5. 21-30
6. 31-40
7. 41+

Poll Question

What type of variable does this survey question give?

(number of hours worked)

1. Continuous
2. Discrete
3. Nominal
4. Ordinal

The variable from the previous slide (the number of hours worked in the last 7 days) is an ordinal variable

Poll Questions

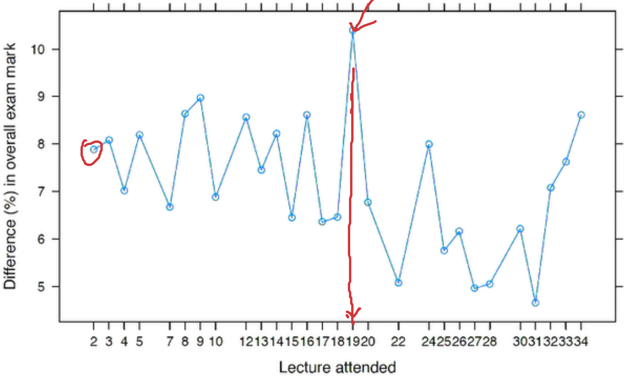
We will use polls in lectures as a way to

1. Test your understanding
2. Highlight tricky concepts
3. Gather quick data for use in examples

Poll Response Data

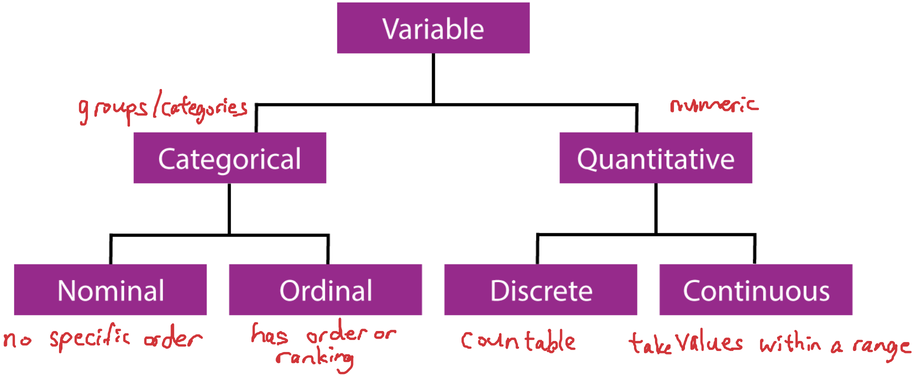
The following plot shows the difference in average exam marks between students who answered poll questions and those who didn't for each of the lectures in 2013:

Which topic was covered in this lecture?



Questions?

Types of Variables



Types of Studies

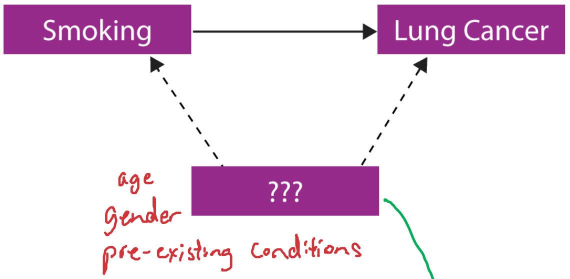
- **Observational Studies**
  - passive
  - difficult to establish causation

no intervention

- **Designed Experiments**
  - designed process of data collection
  - allows for randomisation to reduce bias

intervene with subjects

Randomisation, bias, confounding variables



Randomisation help reduce bias and protect against conofounding variables.

Suspicious?

- Suppose I toss a coin once and it comes up heads. Would you be suspicious about that coin?
- Suppose I toss the coin again and it comes up heads again?
- How many heads in a row would make you suspicious? Poll: 1 to 10

$\mathbb{P}(5 \text{ heads}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32} = 0.03125 \approx 3\%$

typical cutoff used in science: 5%

Alice's Caffeine Experiment

Does caffeinated cola increases pulse rate compared to decaffeinated cola?

Caffeine	Change in Pulse Rate (bpm)									
Yes	17	22	21	16	6	−2	27	15	16	20
No	4	10	7	−9	5	4	5	7	6	12

(See UQ Extend for details.)

1. Measure pulse rate of 20 friends.
2. Give 10 friends 250ml of caf. cola, and the other 10 had 250ml of decaf cola
3. Wait for 30 minutes. Measure pulse rate again. Record the difference in pulse rate.

What are some issues with this design?

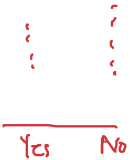
Working with Data in RStudio

Let's use R to compare the mean change in pulse rate between the caffeine group and the caffeine-free group.

$\bar{x} = 15.8$ = mean change for caffeine group

$\bar{y} = 5.1$ = mean change for decaf group

$\bar{x} - \bar{y} = 10.7$ = group difference



stripplot

Poll Question

Does the mean difference of 10.7 bpm give evidence that caffeinated cola increases pulse rate?

1. No
2. Yes, but only weak evidence
3. Yes, moderate evidence
4. Yes, strong evidence

Evidence

Two explanations:

0. Caffeine really has no effect on pulse rate and the observed difference of 10.7 bpm was just **due to the chance variability** (or coincidence) in pulse rates.

1. The difference of 10.7 bpm arose because **caffeine does increase pulse rate**.

How to decide between the two, based on the data?

Randomisation Test

Suppose the first explanation is correct – i.e., there is really no difference.

That is, caffeine has no effect on pulse rate.

Could we get a difference of 10.7 bpm if we shuffled the list of all values of pulse rate changes?

If we could, then that would suggest the labels don't matter and it could just be due to chance.

But if a value like 10.7 bpm is unusual then we would be suspicious of that explanation...

Randomisation Test

How likely is it that we end up with a group difference of 10.7 bpm or more?

1. Put all values (the data) into a list.
2. Shuffle the list. Label the first ten as "Yes", and the next ten as "No". Calculate and record the difference between the two group means.
3. Repeat 2 many times. How often do we get a value ≥ 10.7 ?

Out of 1 million repetitions, only 1950 are 10.7 or bigger.

This gives a proportion of 0.00195.

So, we should be suspiscous that 10.7 is just by chance

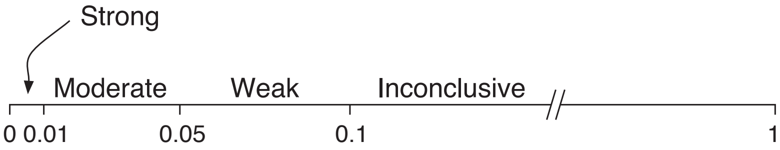
Hypothesis Testing

- The randomisation test is one example of statistical *hypothesis testing*
- The first explanation we gave is called the *null hypothesis*, H_0, of the test ("no association" or "no effect")
- The **probability of obtaining such unusual data**, if indeed H_0 is true, is the p-*value* of the test [here, $p =$ 0.001905]
- We cannot say "the probability that H_0 is true is p"

P-values

- A small p-value suggests H_0 may be wrong, giving evidence for the second explanation
- A large p-value suggests that the data are consistent with the null explanation, giving inconclusive evidence of an effect

Strength of Evidence



Alice's Caffeine Experiment

What do we conclude?

Finding a Paper to Review

Find a paper with a p-value to review:

- No earlier than 2023
- Must have a DOI
- Avoid *retrospective studies* and *meta-analyses*
- Anything with animals or humans will probably be suitable for the Ethics component

Search for topics through databases, such as *PubMed*, or just browse previous papers that have been reviewed and get the latest issue of the journal.

Register through Blackboard.